

Java Output: Numbers

Mathematical operations and numerical display

1. The No-Quotes Rule

Unlike text (Strings), dedicated numeric values are written **without double quotes**. This tells Java to treat them as numbers that can be used for calculations.

Treat as Number:

```
System.out.println(10);
```

Permits math: $10 + 10 = 20$

Treat as Text:

```
System.out.println("10");
```

Permits text join: `"10" + "10" = 1010`

2. Basic Math Operators

Java supports all standard mathematical operators directly inside the print statement.

Operation	Operator	Example	Output
Addition	+	$5 + 3$	8
Subtraction	-	$10 - 4$	6
Multiplication	*	$4 * 3$	12
Division	/	$20 / 5$	4
Modulus (Remainder)	%	$10 \% 3$	1

3. Order of Operations (BODMAS)

Java follows standard mathematics rules. Use **parentheses ()** if you want to force an addition before a multiplication.

```
System.out.println(2 + 3 * 4); // Output: 14
System.out.println((2 + 3) * 4); // Output: 20
```

4. Concatenation vs. Addition

The + sign is versatile. If it touches text, it switches from "Math Mode" to "Join Mode".

The Rule of the Plus (+) Sign:

- Number + Number = **Addition** (10 + 5 → 15)
- Text + Number = **Join** ("Age: " + 10 → Age: 10)
- Text + Text = **Join** ("5" + "5" → 55)

5. Common Pitfalls

- **✗ Quotes around Math:** `System.out.println("5 + 5");` will print literal characters "5 + 5" instead of 10.
- **✗ Division by Zero:** `System.out.println(10 / 0);` will cause your program to crash.
- **✗ Large Integers:** Standard numbers (int) have a limit of roughly +/- 2 billion. For bigger numbers, special types are needed.

6. Practice Challenge

What will the following code display?

```
System.out.println("Result: " + 10 + 5);
```

Answer: Result: 105 (Because once it joins the text "Result: " with 10, the whole thing becomes a String, which then joins with 5!)

Summary Checklist

- Numbers use **NO quotes**.
- Java can calculate directly inside `println()`.
- Multiplication and Division happen **before** Addition.
- The `+` symbol is smart—it knows when to add or join.